installation date on the ship; or

(b) for ships other than those prescribed in (a) above, a contractual delivery date for the equipment or, in the absence of a contractual delivery date, the actual delivery date of the equipment to the ship on or after 1 January 2029.

ANNEX

PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)

1 SCOPE OF ECDIS

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ANNEX

PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)

1 SCOPE OF ECDIS

1.1 The primary function of ECDIS is to contribute to safe navigation.

1.2 ECDIS with adequate backup arrangements may be accepted as complying with the up-to-date charts and nautical publications required by regulations V/19 and V/27 of the 1974 SOLAS Convention. For the purpose of this document, the definition of electronic navigational data service (ENDS) encompasses the nautical charts and nautical publications as defined in SOLAS chapter V and IHO standards in force.

1.3 ECDIS should be capable of displaying all nautical information necessary for safe and efficient navigat ion, originated and distributed by or on the authority of a government , authorized hydrographic office or other relevant government institution, as required by SOLAS regulations V/19 and V/27.

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authorized hydrographic office or other relevant government institution, as required by SOLAS regulations V/19 and V/27.

1.4 ECDIS should facilitate simple and reliable updating of the ENDS.

1.5 ECDIS should reduce the navigational workload compared to using the paper chart and paper nautical publications. It should enable the mariner to execute in a convenient and timely manner all route planning, route monitoring and positioning. It should be capable of continuously indicating, monitoring and recording the ship's position.

1.6 The ECDIS display may also b e used for the display of radar, radar tracked target information, AIS and other appropriate data layers to assist in route monitoring.

1.7 ECDIS should provide appropriate alerts or indications with respect to the information displayed or malfunction of the equipment (see appendix 5). ECDIS should meet the requirements of the Performance standards for bridge alert management (resolution MSC.302(87)).

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requirements of the Performance standards for bridge alert management (resolution MSC.302(87)).

1.8 When the relevant chart information is not available in the appropriate form (see section 4), some ECDIS equipment may operate in the raster chart display system (RCDS) mode as defined in appendix 7. RCDS mode of operation should conform to p erformance standards not inferior to those set out in appendix 7.

2 APPLICATION OF THESE STANDARDS

2.1 These performance standards should apply to all ECDIS equipment carried on all ships, as follows:

.1 dedicated stand-alone workstation; and

,2 a multifunction workstation as part of an INS.

2.2 These performance standards apply to ECDIS mode of operation, ECDIS in RCDS mode of operation as specified in appendix 7 and ECDIS backup arrangements as specified in appendix 6.

RESOLUTION MSC.530(106) (adopted on 7 November 2022)
PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)

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PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)

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2.3 Requirements for structure, format, encryption presentation of the ENDS are within the scope of relevant IHO standards, including those listed in appendix 1.

2.4 In addition to the general requirements set out in resolution A.694(17) 1 and the presentation requirements set out in resolution MSC.191(79), as amended, ECDIS equipment should meet the requirements of these standards and follow the relevant guidelines on ergonomic principles adopted by the Organization.2

3 DEFINITIONS

For the purpose of these performance standards:

3.1 Electronic chart display and information system (ECDIS) means a navigation information system which with adequate backup arrangements can be accepted as complying
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information system which with adequate backup arrangements can be accepted as complying with the up -to-date nautical chart and nautical publications required by SOLAS regulations V/19 and V/27, by displaying selected information from a system database with positional information from navigation sensors to assist the mariner in route planning and route monitoring and, if required, display additional navigation-related information.

3.2 Electronic navigational chart (ENC) means the database, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of a government, authorized hydrographic office or other relevant government institution, and conforming to IHO standards. The ENC contains all the nautical chart information necessary for safe navigation.

3.3 Electronic navigational data service (ENDS) means a special -purpose database compiled from nautical chart and nautical publication data, standardized as to content,
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compiled from nautical chart and nautical publication data, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of a government, authorized hydrographic office or other relevan t government institution, and conforming to IHO standards; and, which is designed to meet the requirement of marine navigation and the nautical charts and nautical publications carriage requirements in SOLAS regulations V/19 and V/27. The navigational base layer of ENDS is the electronic navigational chart (ENC).

3.4 System database means a database, in the manufacturer's internal ECDIS format, resulting from the lossless transformation of the ENDS contents and its updates. It is this database that i s accessed by ECDIS for the display generation and other navigational functions, and is equivalent to up-to-date ENDS.

3.5 Standard display is the display mode intended to be used as a minimum during route
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functions, and is equivalent to up-to-date ENDS.

3.5 Standard display is the display mode intended to be used as a minimum during route planning and route monitoring. The chart content is listed in appendix 2.

3.6 Display base means the chart content as listed in appendix 2 and which cannot be removed from the display. It is not intended to be sufficient for safe navigation.

3.7 Further information on ECDIS definitions may be found in IHO Hydrographic Dictionary Publication S-32 (see appendix 1).

1 MSC/Circ.982.

2 Refer to Publication IEC 60945.
RESOLUTION MSC.530(106) (adopted on 7 November 2022)
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MODULE A DATABASE

4 PROVISION AND UPDATING

4.1 The ENDS information to be used in ECDIS should be issued by or on the authority
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MODULE A DATABASE

4 PROVISION AND UPDATING

4.1 The ENDS information to be used in ECDIS should be issued by or on the authority